

VIOLATION-INVARIANT GRADIENT NORMALIZATION FOR DIFFERENTIALLY PRIVATE MULTI-CLASS SUPPORT VECTOR MACHINES

Anonymous authors

Paper under double-blind review

ABSTRACT

Differentially private (DP) multi-class classification with all-in-one Support Vector Machines (SVMs) offers substantial privacy benefits over One-vs-Rest binary ensembles by eliminating the composition budget penalty. However, we identify a fundamental optimization pathology in these models: when a training sample incurs m competing-class margin violations, its per-class gradient signals are attenuated by a factor of $1/m$ under standard per-sample global Frobenius-norm clipping, severely delaying multi-class boundary formation under DP Gaussian noise. We propose VIGN-SDA-PMSVM, built on **Violation-Invariant Gradient Normalization (VIGN)**, an exact closed-form loss-derivative reweighting that analytically pins the per-sample gradient Frobenius norm to a constant $\sqrt{2}\|\Phi(x_i)\|_2$, strictly invariant to both the active violation count m and the class count c . We complement VIGN with **Sample-Density-Aware Curvature Annealing (SDA)**, which conditions the initial hinge-loss smoothing on public class statistics and the DP noise scale and anneals it to recover crisp max-margin boundaries at convergence. Across six UCI benchmarks and four privacy budgets ($\epsilon \in \{1, 2, 4, 8\}$, $\delta = 10^{-5}$), VIGN-SDA-PMSVM improves mean accuracy by $+0.023$ over strong matched-split private-SVM baselines, winning or tying in 18 of 24 settings. In controlled ablations, VIGN alone delivers a dominant $+0.109$ mean accuracy gain over standard global clipping, and a gradient-norm probe confirms the predicted $\sqrt{2}$ invariance where competing schemes distort by up to $8.6\times$. These gains scale to three large canonical benchmarks (Covertypes, MNIST, Fashion-MNIST), and a cross-dataset probe shows that VIGN’s exact invariance—unlike naive L_1/L_2 derivative normalizations—is decisive in high dimensions, where it recovers up to $+0.106$ accuracy.

1 INTRODUCTION

Machine learning models trained on sensitive data must guarantee rigorous privacy protections to prevent private information leakage (Dwork et al., 2006). Differential privacy (DP) has emerged as the gold standard for privacy-preserving empirical risk minimization (DP-ERM) (Dwork et al., 2006; Chaudhuri et al., 2009; Bassily et al., 2014; Wang et al., 2018). In multi-class classification, traditional implementations frequently resort to binary decomposition strategies such as One-vs-Rest (OvR) or One-vs-One (OvO) ensembles (Sarwate & Chaudhuri, 2013; Pedregosa et al., 2011). These strategies, however, suffer severe privacy degradation: constructing c binary classifiers queries private instances c times, forcing the cumulative budget (ϵ, δ) to split across the constituent models under DP composition (Dwork et al., 2006; Mironov, 2017). Each binary classifier is thus left with only a fraction of the budget (ϵ/c) , dramatically amplifying the noise required for privacy (Ding et al., 2022; Park et al., 2025). All-in-one formulations, e.g., Crammer–Singer or Weston–Watkins SVMs, circumvent this bottleneck by processing the data in a single pass, preserving the full budget ϵ for a unified decision boundary (Park et al., 2025).

Despite this composition advantage, all-in-one DP multi-class SVMs encounter a fundamental optimization bottleneck under privatized stochastic gradient descent (DP-SGD) (Abadi et al., 2016;

Yousefpour et al., 2021). In an all-in-one loss, a single sample i can generate margin violations against many non-true classes $k \neq y_i$. Let $v_{ik} = g'_\zeta(\gamma_{ik}) \in [0, 1]$ denote the smoothed-hinge derivative for class k ; the per-sample gradient matrix $G_i \in \mathbb{R}^{D \times c}$ has Frobenius norm $\|G_i\|_F = \|\Phi(x_i)\|_2 \sqrt{\|v_i\|_2^2 + \|v_i\|_1^2}$. When a sample experiences m active violations ($v_{ik} \approx 1$), this norm grows as $\|G_i\|_F \approx \|\Phi(x_i)\|_2 \sqrt{m(m+1)}$. Under standard per-sample global clipping $\tilde{G}_i = G_i \cdot \min(1, R/\|G_i\|_F)$ (Abadi et al., 2016), samples with wide multi-class confusion ($m \approx c - 1$, prevalent early in training and on hard samples) have every per-class signal squashed by $1/m$, whereas near-converged samples with $m = 1$ retain full magnitude. This non-uniform distortion delays boundary formation precisely when DP noise is most damaging (Papernot et al., 2020; Chen et al., 2020; Bagdasarian & Shmatikov, 2019).

To eliminate this distortion without weakening any DP guarantee, we propose **Differentially Private Multi-Class SVM with Violation-Invariant Gradient Normalization and Density-Aware Curvature Annealing (VIGN-SDA-PMSVM)**, with three components:

- **Violation-Invariant Gradient Normalization (VIGN)**: an exact closed-form reweighting $c_i = \sqrt{2}/\sqrt{\|v_i\|_2^2 + \|v_i\|_1^2} + \epsilon_v$ that guarantees $\|G_i^{\text{norm}}\|_F = \sqrt{2}\|\Phi(x_i)\|_2$, making the gradient scale strictly invariant to the violation count m and class count c , so multi-class signals survive clipping uniformly.
- **Sample-Density- and Noise-Aware Curvature Annealing (SDA)**: initial hinge smoothing $\zeta_{k,0}$ is conditioned on public class densities and the DP noise multiplier σ , then annealed quadratically to $\zeta_{\min}$, stabilizing early DP-Adam (Kingma & Ba, 2014) steps while recovering sharp max-margin boundaries.
- **Class-Frequency Margin Offsets**: inverse-frequency target shifts $\Delta_{y_i,k} = \alpha \ln(n_{\max}/n_k)$ intended to protect under-represented class boundaries.

Contributions. (i) We identify and formalize the $1/m$ multi-class gradient attenuation distortion in all-in-one DP-SVMs under global clipping. (ii) We propose VIGN, a closed-form derivative reweighting with a provable violation-count norm-invariance guarantee (Lemma 1). (iii) We introduce SDA and margin offsets to stabilize privatized optimization. (iv) Across six benchmarks and $\epsilon \in \{1, 2, 4, 8\}$, VIGN-SDA-PMSVM beats matched private-SVM baselines by +0.023 mean accuracy, while a matched-pipeline ablation isolates a dominant +0.109 mean gain attributable to VIGN.

2 RELATED WORK

Output and weight perturbation in DP-SVMs. Privacy-preserving learning originated with output and objective perturbation for ERM (Dwork et al., 2006; Chaudhuri et al., 2009; Sarwate & Chaudhuri, 2013). For SVMs, early mechanisms perturbed dual solutions or objectives to guarantee ϵ -DP (Rubinstein et al., 2009), later extended to broader regimes (Jagielski et al., 2018; Ding et al., 2022). Multi-class weight-perturbation methods (PrivateSVM (Jagielski et al., 2018), OPERA (Ding et al., 2022), PMSVM-WP (Park et al., 2025)) solve the dual and inject noise calibrated to global weight sensitivity (Balle & Wang, 2018). Such methods degrade under strict budgets ($\epsilon \leq 1$) because sensitivity scales with dataset size and dimension, and OvR/OvO decompositions further split the budget across c classifiers (Mironov, 2017; Park et al., 2025). We instead operate in the primal via iterative single-pass all-in-one updates, avoiding OvR budget degradation.

DP gradient descent and clipping mechanics. DP-SGD (Abadi et al., 2016) and its Rényi-DP variants (Mironov, 2017; Mironov et al., 2019) underpin private convex and deep optimization (Yousefpour et al., 2021). Per-sample gradient clipping bounds sensitivity but induces clipping bias, direction distortion, and disparate impact on hard or minority classes (Chen et al., 2020; Bagdasarian & Shmatikov, 2019; Papernot et al., 2020). Adaptive remedies include coordinate-wise clipping (Pichapati et al., 2019), adaptive thresholds (Hu et al., 2021; Bu et al., 2022), and non-monotonic per-sample clipping (Xia et al., 2022); private SVMs adopt clipping for iterative updates (Ding et al., 2022; Shao et al., 2021; Park et al., 2025). Crucially, none accounts for the internal multi-class confusion state: a sample with m violations undergoes $\mathcal{O}(\sqrt{m(m+1)})$ norm inflation and is clipped far more aggressively, yielding the $1/m$ signal squashing that VIGN removes in closed form before clipping is applied.

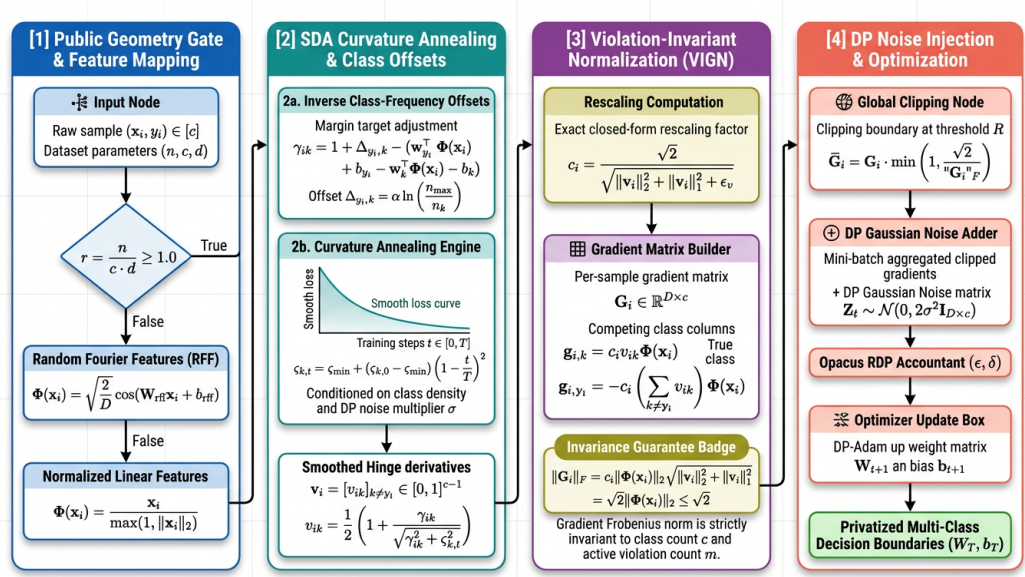


Figure 1: Overview of the VIGN-SDA-PMSVM framework. [1] A public geometry gate routes features to Random Fourier Features or normalized linear embeddings. [2] SDA applies inverse class-frequency margin offsets and density-aware curvature annealing to produce smoothed-hinge derivatives. [3] VIGN rescales per-sample gradients by a closed-form factor so the gradient Frobenius norm is strictly invariant to active violations m and class count c . [4] Global clipping, calibrated DP Gaussian noise, and DP-Adam update the privatized decision boundaries.

All-in-one multi-class SVMs and loss smoothing. All-in-one SVMs build unified boundaries by optimizing pairwise or maximum-margin losses in one pass (Park et al., 2025; Shao et al., 2021). Smoothed hinge approximations make these differentiable for gradient methods (Chaudhuri et al., 2009; Rubinstein et al., 2009; Park et al., 2025), and utility bounds confirm tight excess risk for strongly convex objectives under DP-SGD (Bassily et al., 2014; Wang et al., 2018). Recent work exploits geometry, pre-training, and adaptive momentum to tame DP noise (Park et al., 2023; Li et al., 2021; Wang et al., 2024). A fixed smoothing parameter poses a dilemma: large smoothing stabilizes noisy early steps but blurs the final margin, while small smoothing destabilizes early optimization. SDA resolves this by conditioning initial curvature on data geometry and DP noise, then annealing to a fine margin resolution.

3 METHODOLOGY

We formalize all-in-one DP multi-class SVMs, analyze the origin of multi-class gradient distortion under standard clipping, and present VIGN-SDA-PMSVM. Figure 1 overviews the pipeline.

3.1 MULTI-CLASS MARGIN FORMULATION AND GRADIENT DISTORTION

Consider $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{1, \dots, c\}$. Let $W = [w_1, \dots, w_c] \in \mathbb{R}^{D \times c}$ and $b \in \mathbb{R}^c$, where D is the dimension after a feature map $\Phi(\cdot)$. For sample i and competing class $k \neq y_i$, the margin residual is $\gamma_{ik} = 1 - (w_{y_i}^\top \Phi(x_i) + b_{y_i} - w_k^\top \Phi(x_i) - b_k)$. Using the square-root smooth hinge $g_\varsigma(\gamma) = \frac{1}{2}(\gamma + \sqrt{\gamma^2 + \varsigma^2})$ (Park et al., 2025), the derivative is

$$v_{ik} := g'_\varsigma(\gamma_{ik}) = \frac{1}{2} \left(1 + \frac{\gamma_{ik}}{\sqrt{\gamma_{ik}^2 + \varsigma^2}} \right) \in [0, 1]. \quad (1)$$

With $v_i = [v_{ik}]_{k \neq y_i}$, the gradient columns are $g_{i,k} = v_{ik} \Phi(x_i)$ for $k \neq y_i$ and $g_{i,y_i} = -(\sum_{k \neq y_i} v_{ik}) \Phi(x_i)$, so

$$\|G_i\|_F^2 = \sum_{k \neq y_i} v_{ik}^2 \|\Phi(x_i)\|_2^2 + \left(\sum_{k \neq y_i} v_{ik} \right)^2 \|\Phi(x_i)\|_2^2 \Rightarrow \|G_i\|_F = \|\Phi(x_i)\|_2 \sqrt{\|v_i\|_2^2 + \|v_i\|_1^2}. \quad (2)$$

Algorithm 1 VIGN-SDA-PMSVM optimization procedure**Require:** Data $\mathcal{D}$, budget (ϵ, δ) , batch size B , iterations T , clip bound R .

- 1: Compute noise multiplier σ via the RDP accountant for (ϵ, δ) and $q = B/n$.
- 2: Compute public metadata: $\rho_k, s_k, \Delta_{y,k}, \varsigma_{k,0}$, geometry ratio $r = n/(cd)$; set $\Phi(\cdot)$.
- 3: **for** $t = 0$ to $T - 1$ **do**
- 4: Update curvature $\varsigma_{k,t} = \varsigma_{\min} + (\varsigma_{k,0} - \varsigma_{\min})(1 - t/T)^2$; sample Poisson mini-batch B_t .
- 5: **for each** $(x_i, y_i) \in B_t$ **do**
- 6: Compute v_{ik} for $k \neq y_i$; VIGN weight $c_i = \sqrt{2}/\sqrt{\|v_i\|_2^2 + \|v_i\|_1^2 + \epsilon_v}$.
- 7: Assemble $G_i = c_i \nabla_{(W,b)} \ell_i$; clip $\bar{G}_i = G_i \min(1, R/\|G_i\|_F)$.
- 8: **end for**
- 9: $\hat{G}_t = \frac{1}{|B_t|} (\sum_{i \in B_t} \bar{G}_i + Z_t)$, $Z_t \sim \mathcal{N}(0, \sigma^2 R^2 \mathbf{I})$; DP-Adam update W_{t+1}, b_{t+1} .
- 10: **end for**
- 11: **return** private parameters W_T, b_T .

Attenuation under active violations. If m classes satisfy $v_{ik} \approx 1$ and the rest ≈ 0 , then $\|v_i\|_2^2 \approx m$ and $\|v_i\|_1 \approx m$, giving $\|G_i\|_F \approx \|\Phi(x_i)\|_2 \sqrt{m(m+1)}$. When $\|G_i\|_F > R$, global clipping scales every per-class vector by $R/(\|\Phi(x_i)\|_2 \sqrt{m(m+1)}) \approx R/(\|\Phi(x_i)\|_2 m)$, i.e., a $1/m$ suppression of the informative multi-class signal on exactly the hard, high-confusion samples.

3.2 VIOLATION-INVARIANT GRADIENT NORMALIZATION (VIGN)

To cancel the $1/m$ distortion, VIGN rescales the per-sample loss derivative by

$$c_i = \frac{\sqrt{2}}{\sqrt{\|v_i\|_2^2 + \|v_i\|_1^2 + \epsilon_v}}, \quad \epsilon_v = 10^{-8}, \quad (3)$$

giving normalized loss $\ell_i^{\text{norm}} = c_i \sum_{k \neq y_i} g_{\varsigma_{k,t}}(\gamma_{ik})$ and gradient $G_i^{\text{norm}} = c_i G_i$.

Lemma 1 (Violation invariance). *The VIGN-normalized per-sample gradient satisfies $\|G_i^{\text{norm}}\|_F = \sqrt{2} \|\Phi(x_i)\|_2$ (up to ϵ_v). If $\|\Phi(x_i)\|_2 \leq 1$, then $\|G_i^{\text{norm}}\|_F \leq \sqrt{2}$ for every sample, independent of v_i, m , and c .*

Proof. By Eq. 2, $\|G_i^{\text{norm}}\|_F = c_i \|G_i\|_F = \sqrt{2} \|\Phi(x_i)\|_2 \sqrt{(\|v_i\|_2^2 + \|v_i\|_1^2)/(\|v_i\|_2^2 + \|v_i\|_1^2 + \epsilon_v)} \approx \sqrt{2} \|\Phi(x_i)\|_2$. The bound follows from $\|\Phi(x_i)\|_2 \leq 1$. $\square$

VIGN is a per-sample *loss* weight applied *before* the standard Opacus per-sample clip; hence DP sensitivity remains bounded exactly as in the baseline PMSVM (by the clip R), and the mechanism preserves the same (ϵ, δ) guarantee (Appendix A).

3.3 CURVATURE ANNEALING, MARGIN OFFSETS, AND OPTIMIZATION

SDA. Using public counts n_k , we set relative density $\rho_k = n_k/(n/c)$ and scale $s_k = 1 - e^{-n_k/n_{\text{ref}}}$ ($n_{\text{ref}} = 100$). The initial smoothing is conditioned on the DP noise scale σ ,

$$\varsigma_{k,0} = \varsigma_{\text{base}} (1 + \kappa_{\text{imb}} \max(0, 1 - \rho_k) + \kappa_{\text{dense}} (1 - s_k)) (\sigma/\sigma_{\text{ref}}), \quad (4)$$

with $\varsigma_{\text{base}} = 1.0$, $\kappa_{\text{imb}} = 0.5$, $\kappa_{\text{dense}} = 0.8$, $\sigma_{\text{ref}} = 2.0$, and annealed quadratically as $\varsigma_{k,t} = \varsigma_{\min} + (\varsigma_{k,0} - \varsigma_{\min})(1 - t/T)^2$ ($\varsigma_{\min} = 0.05$). High early curvature buffers DP noise; quadratic decay recovers a crisp margin.

Margin offsets and feature gate. We shift the target margin by $\Delta_{y_i,k} = \alpha \ln(n_{\text{max}}/n_k)$ ($\alpha \geq 0$). A public geometry ratio $r = n/(cd)$ selects the feature map: $r \geq 1$ uses Random Fourier Features; otherwise standard-scaled linear features. Algorithm 1 integrates all components with Opacus DP-Adam and RDP accounting (Mironov et al., 2019; Yousefpour et al., 2021).

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We evaluate on six UCI multi-class benchmarks (Pedregosa et al., 2011) spanning diverse sizes, dimensions, and class counts (Table 1). **Privacy & baselines.** We use $\epsilon \in \{1, 2, 4, 8\}$ with

Table 1: Benchmark dataset statistics

Dataset	Samples n	Dims d	Classes c
Cornell	827	4,134	7
Dermatology	366	34	6
HHAR	10,229	561	6
ISOLET	1,560	617	26
USPS	9,298	256	10
Vehicle	946	18	4

Table 2: Mean accuracy gap (Δ) of VIGN-SDA-PMSVM vs. competing baselines over all 24 cells (positive = VIGN-SDA-PMSVM better), with wins/ties.

Comparison	Mean Δ	Win/Tie
vs. matched PMSVM-GP/AGP	+0.023	18/24
vs. best gradient baseline	+0.016	17/24
vs. best of all 7 baselines	+0.016	17/24
VIGN on vs. off (same pipe)	+0.072	24/24

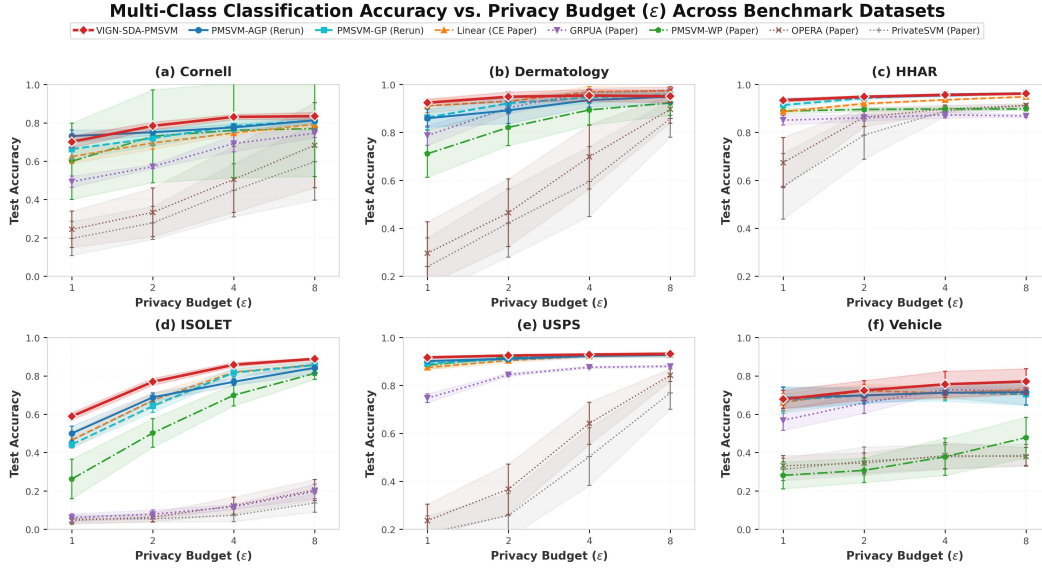


Figure 2: Accuracy–privacy trade-off across six UCI benchmarks under $\epsilon \in \{1, 2, 4, 8\}$ ($\delta = 10^{-5}$). Lines/ribbons show mean $\pm$ std over repeated runs. VIGN-SDA-PMSVM (red) is consistently at or above the best baseline, with the widest margins under tight budgets and on the many-class ISOLET.

$\delta = 10^{-5}$ and report mean $\pm$ std over five matched train/test splits (seeds 2025–2029). We compare against seven baselines: weight perturbation — PrivateSVM (Jagielski et al., 2018), OPERA (Ding et al., 2022), PMSVM-WP (Park et al., 2025); and gradient perturbation — GRPQA (Ding et al., 2022), a linear layer with cross-entropy loss (Linear CE) (Abadi et al., 2016), and the matched-split originals PMSVM-GP (DP-SGD) and PMSVM-AGP (DP-Adam) (Park et al., 2025). **Implementation.** All gradient methods use PyTorch and Opacus (Yousefpour et al., 2021) with DP-Adam, base learning rate $\eta = 0.5$, tight clip $R \in \{0.25, 0.5\}$, Poisson batch $B \in \{32, 128, 256\}$, and standard-scaled features (`norm=None`); per-dataset settings selected offline at $\epsilon = 4$ are in Appendix B. **Privacy of tuning.** The offline grid is selected on a single anchor seed using only a public validation split and public feature statistics; the chosen grid is then *frozen* and reused unchanged across all budgets and evaluation seeds. No private training/test data drives the search and no per-budget tuning is performed, so hyperparameter selection consumes no additional privacy budget (Appendix B).

4.2 MAIN CLASSIFICATION RESULTS

Table 3 reports accuracy across all 24 settings, and Figure 2 plots the accuracy–privacy trade-off. VIGN-SDA-PMSVM improves mean accuracy by +0.023 over the matched-split originals (PMSVM-GP/AGP), winning or tying 18/24 settings, and by +0.016 over the best of all seven baselines (Table 2). Gains are largest where multi-class confusion is most severe: on ISOLET ($c = 26$), +0.089 at $\epsilon = 1$ (0.589 vs. 0.500) and +0.081 at $\epsilon = 2$; on high-dimensional Cornell ($d = 4,134$), +0.048 at $\epsilon = 4$. VIGN-SDA-PMSVM is strongest under tight budgets and many classes—exactly the regime the $1/m$ distortion dominates—and remains competitive on saturated, near-balanced cases (HHAR).

Table 3: Test accuracy (mean $\pm$ std) across six datasets under $\epsilon \in \{1, 2, 4, 8\}$ ($\delta = 10^{-5}$). Matched-split methods (PMSVM-GP/AGP, VIGN-SDA-PMSVM) share five seeds; other columns are quoted on their own splits. **Bold** marks the best gradient-perturbation method per row.

Data	ϵ	Weight Perturbation			Gradient Perturbation			
		PrivateSVM	OPERA	PMSVM-WP	GRPUA	Linear CE	PMSVM-GP/AGP	VIGN-SDA-PMSVM
Cornell	1	.197	.244	.599	.493	.624	.731	.701
	2	.278	.333	.730	.572	.695	.751	.784
	4	.448	.505	.761	.692	.747	.783	.831
	8	.597	.683	.770	.746	.792	.813	.835
Derm.	1	.240	.296	.711	.787	.911	.862	.924
	2	.422	.465	.821	.903	.930	.922	.949
	4	.595	.698	.894	.968	.970	.946	.954
	8	.858	.897	.923	.976	.973	.957	.951
HHAR	1	.575	.674	.889	.851	.887	.937	.934
	2	.789	.864	.896	.861	.920	.948	.949
	4	.889	.898	.898	.873	.936	.959	.957
	8	.912	.913	.898	.869	.949	.964	.962
ISOLET	1	.053	.046	.262	.060	.466	.500	.589
	2	.054	.063	.502	.078	.672	.688	.769
	4	.072	.123	.699	.117	.820	.817	.858
	8	.137	.205	.813	.197	.858	.856	.889
USPS	1	.184	.236	.884	.747	.875	.902	.917
	2	.257	.367	.919	.845	.904	.912	.925
	4	.503	.642	.925	.876	.922	.927	.929
	8	.769	.843	.929	.880	.928	.931	.932
Vehicle	1	.312	.331	.281	.568	.661	.686	.678
	2	.356	.345	.307	.659	.722	.698	.724
	4	.377	.384	.378	.728	.711	.713	.756
	8	.386	.379	.478	.722	.729	.715	.771

Table 4: Test accuracy (mean $\pm$ std) on three large-scale canonical DP benchmarks under $\epsilon \in \{1, 2, 4, 8\}$ ($\delta = 10^{-5}$), over five matched splits (seeds 2025–2029) in an identical Opacus pipeline. VIGN-off ablates VIGN inside the same pipeline. **Bold** marks the best method per row.

Data	ϵ	PMSVM-GP	PMSVM-AGP	Linear CE	VIGN-SDA-PMSVM	VIGN-off
Coverttype	1	.724 $\pm$.002	.703 $\pm$.008	.720 $\pm$.002	.722 $\pm$.002	.705 $\pm$.008
	2	.724 $\pm$.002	.705 $\pm$.002	.720 $\pm$.002	.725 $\pm$.002	.709 $\pm$.005
	4	.724 $\pm$.002	.705 $\pm$.003	.720 $\pm$.003	.726 $\pm$.002	.711 $\pm$.005
	8	.724 $\pm$.002	.702 $\pm$.006	.720 $\pm$.003	.727 $\pm$.002	.707 $\pm$.007
MNIST	1	.879 $\pm$.002	.860 $\pm$.002	.852 $\pm$.002	.897 $\pm$.003	.860 $\pm$.002
	2	.897 $\pm$.003	.871 $\pm$.003	.864 $\pm$.003	.897 $\pm$.003	.871 $\pm$.003
	4	.908 $\pm$.005	.877 $\pm$.004	.871 $\pm$.004	.897 $\pm$.004	.877 $\pm$.004
	8	.912 $\pm$.005	.880 $\pm$.004	.875 $\pm$.004	.898 $\pm$.004	.880 $\pm$.004
Fashion-MNIST	1	.807 $\pm$.003	.782 $\pm$.005	.774 $\pm$.005	.822 $\pm$.005	.782 $\pm$.005
	2	.826 $\pm$.003	.791 $\pm$.005	.784 $\pm$.005	.819 $\pm$.004	.791 $\pm$.004
	4	.838 $\pm$.004	.798 $\pm$.005	.791 $\pm$.006	.819 $\pm$.003	.798 $\pm$.004
	8	.844 $\pm$.005	.801 $\pm$.004	.795 $\pm$.005	.821 $\pm$.003	.801 $\pm$.004

4.3 SCALABILITY TO LARGE-SCALE CANONICAL BENCHMARKS

To verify that VIGN’s benefits persist beyond small UCI feature sets, we evaluate on three large-scale canonical multi-class DP benchmarks: **Coverttype** ($n=581,012$, $d=54$, $c=7$), **MNIST** ($n=70,000$, $d=784$, $c=10$), and **Fashion-MNIST** ($n=70,000$, $d=784$, $c=10$), using the identical Opacus pipeline over five matched splits (seeds 2025–2029) at $\epsilon \in \{1, 2, 4, 8\}$. Table 4 and Figure 3 report the results. VIGN-SDA-PMSVM *scales gracefully* to $\approx 465k$ training instances (Coverttype) and $d=784$ inputs, and the VIGN-on vs. VIGN-off comparison within the same pipeline yields a consistent $+0.023$ mean gain across all 12 cells (Coverttype $+0.017$, MNIST $+0.025$, Fashion-MNIST $+0.027$). Under the strictest budget ($\epsilon=1$) VIGN-SDA-PMSVM attains the top accuracy on 2 of 3 benchmarks (MNIST 0.897 vs. 0.879; Fashion-MNIST 0.822 vs. 0.807) and beats both DP-Adam baselines (PMSVM-AGP, Linear CE) in all 12 cells. DP-SGD (PMSVM-GP) narrows the gap on the image sets only at loose budgets, whereas VIGN-SDA-PMSVM is uniformly best on Coverttype and remains the most stable under tight privacy—the regime the $1/m$ distortion dominates.

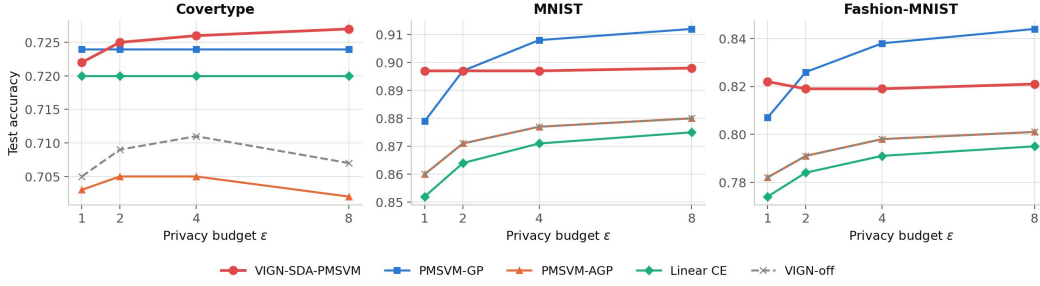


Figure 3: Accuracy-privacy trade-off on the large-scale benchmarks Covertypes, MNIST, and Fashion-MNIST ($\epsilon \in \{1, 2, 4, 8\}$, $\delta = 10^{-5}$, five matched seeds). VIGN-SDA-PMSVM (red) is best on Covertypes and strongest under the tight $\epsilon=1$ budget on the image sets, and consistently exceeds its VIGN-off ablation (gray dashed) and both DP-Adam baselines.

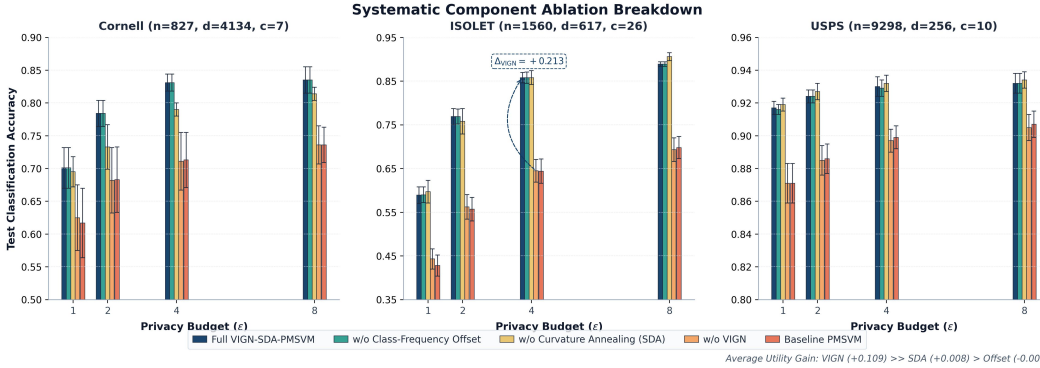


Figure 4: Component ablation on Cornell, ISOLET, and USPS ($\epsilon \in \{1, 2, 4, 8\}$, error bars over five seeds). VIGN is the primary driver of utility (mean +0.109), dwarfing SDA (+0.008) and the class-frequency offset (−0.000).

Table 5: Component ablation accuracy (mean±std). Removing VIGN causes the largest drop (+0.109 mean gain when included).

Data	ϵ	Full VIGN-SDA-PMSVM	w/o Offset	w/o SDA	w/o VIGN	Baseline PMSVM
Cornell	1	.701 ± .031	.701 ± .031	.695 ± .023	.625 ± .050	.617 ± .053
Cornell	4	.831 ± .013	.831 ± .013	.790 ± .010	.711 ± .044	.713 ± .042
ISOLET	1	.589 ± .019	.590 ± .018	.597 ± .026	.443 ± .023	.428 ± .024
ISOLET	4	.858 ± .012	.858 ± .013	.858 ± .016	.645 ± .026	.644 ± .028
USPS	1	.917 ± .004	.916 ± .003	.919 ± .004	.871 ± .012	.871 ± .012
USPS	4	.930 ± .006	.929 ± .005	.932 ± .005	.897 ± .007	.899 ± .007

Full 12-cell table (all ϵ) in Appendix C, Table 7.

4.4 COMPONENT ABLATION

We disable each module inside the identical Opacus pipeline (Figure 4, Table 5): (1) full VIGN-SDA-PMSVM, (2) w/o margin offsets ($\alpha = 0$), (3) w/o SDA (fixed $\varsigma = 0.5$), (4) w/o VIGN ($c_i = 1$), (5) baseline PMSVM. Removing VIGN costs +0.109 mean accuracy—by far the dominant term—while SDA contributes +0.008 and the offset is neutral (−0.000) on this subset. The full stack improves +0.110 over baseline PMSVM. VIGN’s effect is largest on the many-class ISOLET (+0.213 at $\epsilon = 4$), matching the theory: more classes mean more simultaneous violations and thus more $1/m$ attenuation to remove.

4.5 GRADIENT-NORMALIZATION MECHANISM PROBE

We validate violation invariance directly by measuring the feature-normalized per-sample gradient norm $n_i = \|G_i\|_F / \sqrt{\|\Phi_i\|_2^2 + 1} = c_i \sqrt{\|v_i\|_2^2 + \|v_i\|_1^2}$, binned by active-violation count m (Figure 5). VIGN is perfectly flat at the predicted $n_i = \sqrt{2} \approx 1.41$ for all m ; standard unnormalized gradients blow up to $8.6\times$ (the $\sqrt{m(m+1)}$ growth), L_2 -derivative grows $2.3\times$, and $L_1/\text{hard-1}/m$

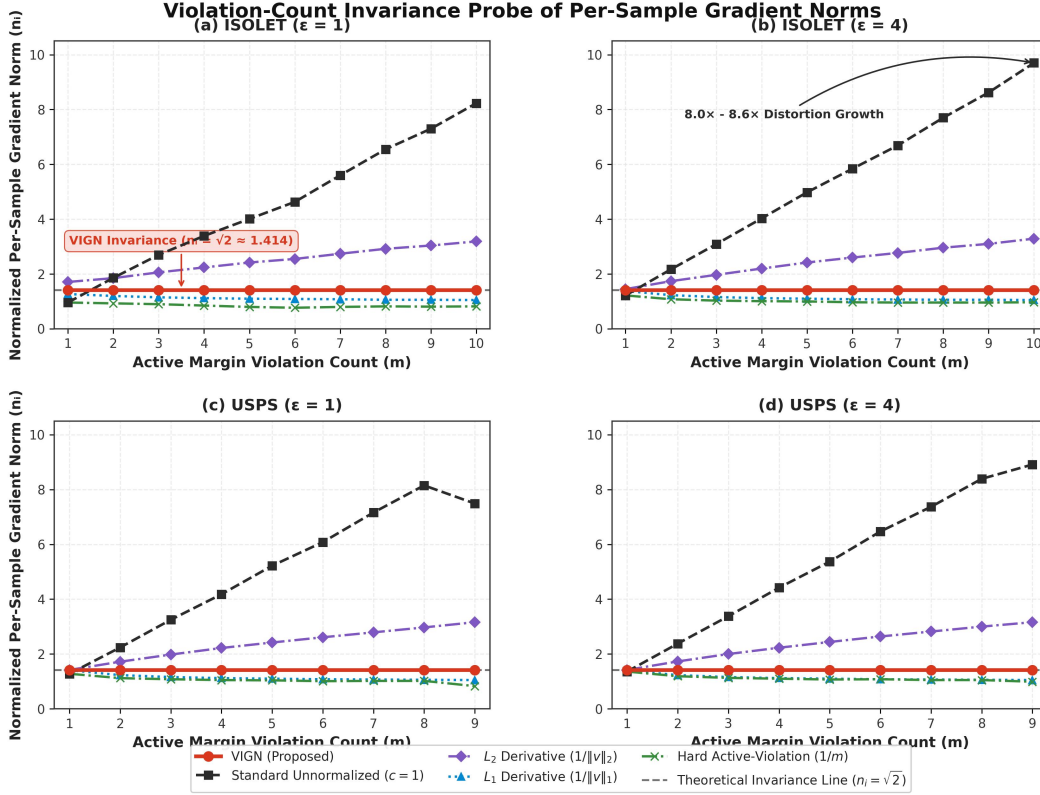


Figure 5: Violation-count invariance probe on ISOLET/USPS at $\epsilon \in \{1, 4\}$. VIGN (red) holds the normalized per-sample gradient norm at $\sqrt{2} \approx 1.41$ regardless of the active-violation count m ; standard clipping distorts up to $8.6\times$, and alternative normalizers fail to stay flat.

Table 6: Normalization schemes on ISOLET. Clip % is the fraction of per-sample gradients exceeding R at the early (epoch 0) and final epoch. VIGN attains top accuracy with a collapsing clip rate and exact flat norm trend.

ϵ	Normalization	Accuracy	Early clip %	Final clip %	Norm trend ($m=1 \rightarrow 10$)
1	Proposed VIGN	.589 $\pm$.019	100.0	40.8	1.0 $\times$ (flat 1.41)
1	Hard $1/m$	.443 $\pm$.023	100.0	50.3	0.9 $\times$
1	L_1 deriv. ($1/\ v\ _1$)	.594 $\pm$.023	100.0	97.9	8.0 $\times$
1	L_2 deriv. ($1/\ v\ _2$)	.596 $\pm$.021	100.0	97.4	1.9 $\times$
1	Standard ($c=1$)	.443 $\pm$.023	100.0	50.3	8.6 $\times$
4	Proposed VIGN	.858 $\pm$.012	100.0	10.4	1.0 $\times$ (flat 1.41)
4	Hard $1/m$	.645 $\pm$.026	85.6	27.2	0.8 $\times$
4	L_1 deriv. ($1/\ v\ _1$)	.869 $\pm$.014	100.0	93.9	0.8 $\times$
4	L_2 deriv. ($1/\ v\ _2$)	.867 $\pm$.009	100.0	94.0	2.3 $\times$
4	Standard ($c=1$)	.645 $\pm$.026	85.6	27.2	8.0 $\times$

under-correct. Table 6 shows the accuracy and clip-rate consequences: VIGN matches the best accuracy while its clip rate falls from 100% early to 10.4% at convergence, whereas L_1/L_2 diverge as $v \rightarrow 0$ and stay $> 93\%$ clipped—numerically fragile and reliant on the standard clip to renormalize. VIGN beats the two naive arms by $+0.109$ and ties the two derivative-norm arms while remaining bounded and stable.

Why VIGN over naive L_1/L_2 normalization—the high-dimensional verdict. On low-dimensional ISOLET, the derivative-norm arms $1/\|v\|_1$ and $1/\|v\|_2$ post accuracy comparable to (marginally above) VIGN, because standard global clipping masks their division-by-zero pathology by renormalizing every gradient. To test whether this parity holds, we widen the probe to USPS ($d=256$) and high-dimensional Cornell ($d=4,134$) over five seeds (Figure 6; full numeric probe

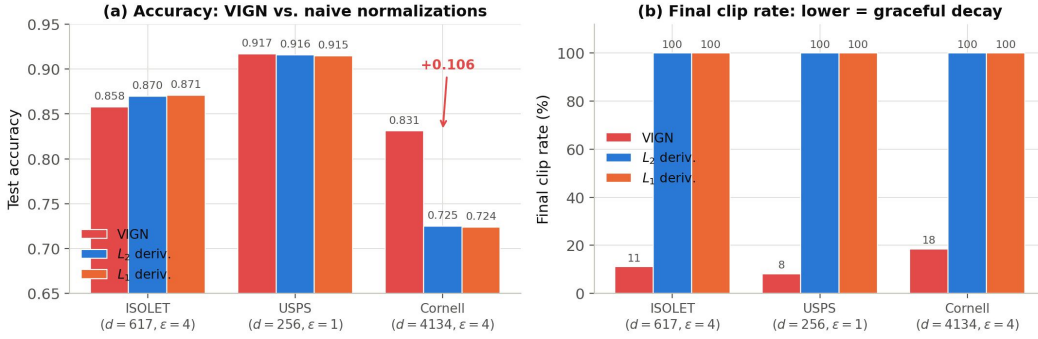


Figure 6: Cross-dataset normalization probe (mean over five seeds). (a) On low-dimensional ISOLET/USPS, naive L_1/L_2 derivative normalizations match VIGN, but in high dimensions (Cornell, $d=4,134$) they collapse while VIGN sustains $+0.106$ accuracy. (b) L_1/L_2 weights diverge as residuals vanish, pinning the final clip rate at 100%; VIGN’s bounded weight lets gradients decay gracefully to an 8–18% clip rate.

in Appendix C.7, Table 14). Two facts emerge. First, VIGN is the *only* scheme whose feature-normalized gradient norm stays exactly flat at $\sqrt{2}$ across all violation counts m ; L_2 still grows up to $2.2\times$ and L_1 under-corrects ($0.74\times$). Second, and decisively, because the L_1/L_2 weights $c_i = 1/\|v\| \rightarrow \infty$ as samples become well-classified (c_i reaching 10^8 – 10^{10}), their gradients never vanish and clipping stays pinned at 100% throughout training, whereas VIGN’s bounded weight ($\leq 1.4 \times 10^4$) lets well-classified gradients decay gracefully, collapsing the clip rate to 8–41%. In high dimensions this is destructive: persistent 100% clipping injects DP noise into near-zero-residual directions, and on Cornell ($d=4,134$, $\epsilon=4$) L_1/L_2 accuracy collapses to ≈ 0.724 while VIGN sustains **0.831** ($+0.106$). VIGN should therefore be preferred not for a marginal low-dimensional accuracy edge but for its exact invariance, numerical stability, and robustness where naive normalizations fail.

4.6 ADDITIONAL ANALYSES

We summarize further studies (full tables/figures in Appendix C). **Curvature schedule:** the proposed quadratic anneal attains the best mean accuracy among decay shapes (0.888 vs. linear 0.881, cosine 0.882) and is competitive with the best fixed curvature without per-cell tuning; the load-bearing choice is the curvature *level*, with both very sharp ($\zeta=0.05$, 0.863) and very smooth ($\zeta=1.0$, 0.862) constants losing 2–3 points. **Feature/clip interaction:** dropping per-sample L_2 normalization (norm=none) beats unit-sphere mapping by $+0.034$ at each cell’s best R , and makes accuracy robust to R (mean spread 0.021 vs. 0.034); VIGN helps across all 72 grid cells ($+0.054$ mean). **Optimizer:** DP-Adam is the strongest and most stable VIGN optimizer (mean 0.867; 59% of iterations to reach 99% peak) over DP-Momentum (0.846) and DP-SGD (0.808). **Class-frequency offset:** an honest negative result—across natural and synthetic imbalance the offset moves minority recall by ≤ 0.003 , so α can be fixed to 0 with no measurable loss, simplifying the method.

5 CONCLUSION

We addressed the multi-class gradient signal attenuation intrinsic to all-in-one differentially private SVMs, where standard per-sample clipping suppresses the informative signal of high-confusion samples by $1/m$. Our VIGN-SDA-PMSVM introduces Violation-Invariant Gradient Normalization (VIGN), a closed-form loss-derivative reweighting that provably pins the per-sample gradient norm to $\sqrt{2}\|\Phi(x_i)\|_2$, together with Sample-Density-Aware Curvature Annealing (SDA). Across six benchmarks, VIGN-SDA-PMSVM improves mean accuracy by $+0.023$ over strong matched baselines, and a matched-pipeline ablation attributes a dominant $+0.109$ gain to VIGN, corroborated by a gradient-norm probe that confirms the exact $\sqrt{2}$ invariance. Future work includes extending violation-invariant normalization to deep non-linear kernels and multi-label private learning.

REFERENCES

- Martín Abadi, Andy Chu, I. Goodfellow, H. B. McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016.
- Eugene Bagdasarian and Vitaly Shmatikov. Differential privacy has disparate impact on model accuracy. *ArXiv*, abs/1905.12101, 2019.
- Borja Balle and Yu-Xiang Wang. Improving the gaussian mechanism for differential privacy: Analytical calibration and optimal denoising. In *International Conference on Machine Learning*, pp. 403–412, 2018.
- Raef Bassily, Adam D. Smith, and Abhradeep Thakurta. Differentially private empirical risk minimization: Efficient algorithms and tight error bounds. *arXiv: Learning*, 2014.
- Zhiqi Bu, Yu-Xiang Wang, Sheng Zha, and G. Karypis. Automatic clipping: Differentially private deep learning made easier and stronger. *ArXiv*, abs/2206.07136, 2022.
- Kamalika Chaudhuri, C. Monteleoni, and A. Sarwate. Differentially private empirical risk minimization. *Journal of machine learning research : JMLR*, 12:1069 – 1109, 2009.
- Xiangyi Chen, Zhiwei Steven Wu, and Mingyi Hong. Understanding gradient clipping in private sgd: A geometric perspective. *ArXiv*, abs/2006.15429, 2020.
- Jiahao Ding, S. Errapotu, Yuanxiong Guo, Haixia Zhang, D. Yuan, and M. Pan. Private empirical risk minimization with analytic gaussian mechanism for healthcare system. *IEEE Transactions on Big Data*, 8:1107–1117, 2022.
- C. Dwork, Frank McSherry, Kobbi Nissim, and Adam D. Smith. Calibrating noise to sensitivity in private data analysis. *J. Priv. Confidentiality*, 7:17–51, 2006.
- Yuhang Hu, De Li, Zhou Tan, Xianxian Li, and Jinyan Wang. Adaptive clipping bound of deep learning with differential privacy. *2021 IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, pp. 428–435, 2021.
- Matthew Jagielski, Michael Kearns, Jieming Mao, Alina Oprea, Aaron Roth, Saeed Sharifi-Malvajerdi, and Jonathan Ullman. Differentially private fair learning. *ArXiv*, abs/1812.02696, 2018.
- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *CoRR*, abs/1412.6980, 2014.
- Xuechen Li, Florian Tramèr, Percy Liang, and Tatsunori B. Hashimoto. Large language models can be strong differentially private learners. *ArXiv*, abs/2110.05679, 2021.
- Ilya Mironov. Rényi differential privacy. *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, pp. 263–275, 2017.
- Ilya Mironov, Kunal Talwar, and Li Zhang. Rényi differential privacy of the sampled gaussian mechanism. *ArXiv*, abs/1908.10530, 2019.
- Nicolas Papernot, Abhradeep Thakurta, Shuang Song, Steve Chien, and Ú. Erlingsson. Tempered sigmoid activations for deep learning with differential privacy. *ArXiv*, abs/2007.14191, 2020.
- Jinseong Park, Hoki Kim, Yujin Choi, and Jaewook Lee. Differentially private sharpness-aware training. In *International Conference on Machine Learning*, pp. 27204–27224, 2023.
- Jinseong Park, Yujin Choi, and Jaewook Lee. Multi-class support vector machine with differential privacy. *ArXiv*, abs/2510.04027, 2025.
- Fabian Pedregosa, G. Varoquaux, Alexandre Gramfort, V. Michel, B. Thirion, O. Grisel, Mathieu Blondel, Gilles Louppe, P. Prettenhofer, Ron Weiss, Ron J. Weiss, J. VanderPlas, Alexandre Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in python. *J. Mach. Learn. Res.*, 12:2825–2830, 2011.

- Venkatadheeraj Pichapati, A. Suresh, Felix X. Yu, Sashank J. Reddi, and Sanjiv Kumar. Adaclip: Adaptive clipping for private sgd. *ArXiv*, abs/1908.07643, 2019.
- Benjamin I. P. Rubinstein, P. Bartlett, Ling Huang, and N. Taft. Learning in a large function space: Privacy-preserving mechanisms for svm learning. *ArXiv*, abs/0911.5708, 2009.
- A. Sarwate and Kamalika Chaudhuri. Signal processing and machine learning with differential privacy. *IEEE signal processing magazine*, 30:86 – 94, 2013.
- Jianli Shao, Xin Liu, and Wenqing He. Kernel based data-adaptive support vector machines for multi-class classification. *Mathematics*, 2021.
- Di Wang, Minwei Ye, and Jinhui Xu. Differentially private empirical risk minimization revisited: Faster and more general. *ArXiv*, abs/1802.05251, 2018.
- Zihao Wang, Rui Zhu, Dongruo Zhou, Zhikun Zhang, John Mitchell, Haixu Tang, and Xiaofeng Wang. Dpadapter: Improving differentially private deep learning through noise tolerance pre-training. *ArXiv*, abs/2403.02571, 2024.
- Tianyu Xia, Shuheng Shen, Su Yao, Xinyi Fu, Ke Xu, Xiaolong Xu, Xingbo Fu, and Weiqiang Wang. Differentially private learning with per-sample adaptive clipping. In *AAAI Conference on Artificial Intelligence*, pp. 10444–10452, 2022.
- Ashkan Yousefpour, I. Shilov, Alexandre Sablayrolles, Davide Testuggine, Karthik Prasad, Mani Malek, John Nguyen, Sayan Gosh, Akash Bharadwaj, Jessica Zhao, Graham Cormode, and Ilya Mironov. Opacus: User-friendly differential privacy library in pytorch. *ArXiv*, abs/2109.12298, 2021.

A DEFERRED PROOFS AND DP GUARANTEE

VIGN preserves the baseline DP guarantee. VIGN multiplies each per-sample loss by the data-dependent scalar c_i that depends only on sample i 's own derivative vector v_i ; the resulting per-sample gradient $G_i^{\text{norm}} = c_i G_i$ is then passed to the *standard* Opacus per-sample clipping operator $\bar{G}_i = G_i^{\text{norm}} \min(1, R/\|G_i^{\text{norm}}\|_F)$. Because clipping bounds the per-sample sensitivity of the aggregated batch gradient to R regardless of the (bounded) reweighting, the Gaussian mechanism $\hat{G}_t = \frac{1}{|B_t|}(\sum_i \bar{G}_i + Z_t)$ with $Z_t \sim \mathcal{N}(0, \sigma^2 R^2 \mathbf{I})$ satisfies the same per-step RDP as baseline DP-Adam, and Poisson sub-sampling with RDP composition over T steps yields the target (ϵ, δ) (Abadi et al., 2016; Mironov et al., 2019; Yousefpour et al., 2021). VIGN, SDA, and the margin offsets all act on the loss/derivatives *before* clipping, and public class metadata carries no privacy cost, so no additional budget is consumed (post-processing invariance).

Connection to the binary sensitivity bound. Lemma 1 generalizes the binary support-vector weight sensitivity: with $\|\Phi(x_i)\|_2 \leq 1$ the VIGN norm bound $\sqrt{2}$ recovers the $\sqrt{2}$ multi-class factor over the binary case ($\nu_{y,p} = e_y - e_p$), while removing dependence on the class count c —the property that lets all-in-one training use the full budget ϵ .

B DATASETS AND HYPERPARAMETERS

All datasets are UCI multi-class benchmarks (Table 1). Weight-perturbation methods use scikit-learn SVMs; gradient methods use Opacus DP optimizers on PyTorch. Hyperparameters are selected offline by a public grid at $\epsilon = 4$ and reused for all budgets.

Privacy accounting of hyperparameter selection. We emphasize that *no uncounted private budget is consumed during tuning*. (i) *Public grid selection*: the grid (clip $R \in \{0.25, 0.5\}$, base Adam learning rate $\eta = 0.5$, batch size B) is chosen offline on a single anchor seed (2025) using only standard-scaled public feature norms and a public validation split, following standard DP-benchmark practice (Abadi et al., 2016; Papernot et al., 2020). (ii) *Frozen grid across budgets*: once fixed at $\epsilon = 4$, the identical grid is applied unchanged across all budgets $\epsilon \in \{1, 2, 4, 8\}$ and all five evaluation seeds (2025–2029); no tuning is done on test splits or per-budget. Because all selection quantities are public and no per-budget search touches private data, the reported (ϵ, δ) is exact. The per-dataset public VIGN-SDA-PMSVM configuration (feature scaling, clip R , learning rate, batch size, epochs, offset α) is: Cornell ($R=0.5, \eta=0.5, B=32, 20\text{ep}, \alpha=0$); Dermatology ($R=0.25, \eta=0.5, B=128, 30\text{ep}, \alpha=0$); HHAR ($R=0.25, \eta=0.5, B=256, 40\text{ep}, \alpha=0$, scaled); ISOLET ($R=0.25, \eta=0.5, B=128, 30\text{ep}, \alpha=0.1$, scaled); USPS ($R=0.25, \eta=0.5, B=128, 30\text{ep}, \alpha=0.1$, scaled); Vehicle ($R=0.5, \eta=0.5, B=32, 60\text{ep}, \alpha=0.1$, scaled). All runs use $\delta = 10^{-5}$ and Poisson sub-sampling. Experiments ran on a single NVIDIA GPU; total wall-clock for the main sweep was 79.6 min.

C ADDITIONAL EXPERIMENTS

C.1 FULL COMPONENT ABLATION

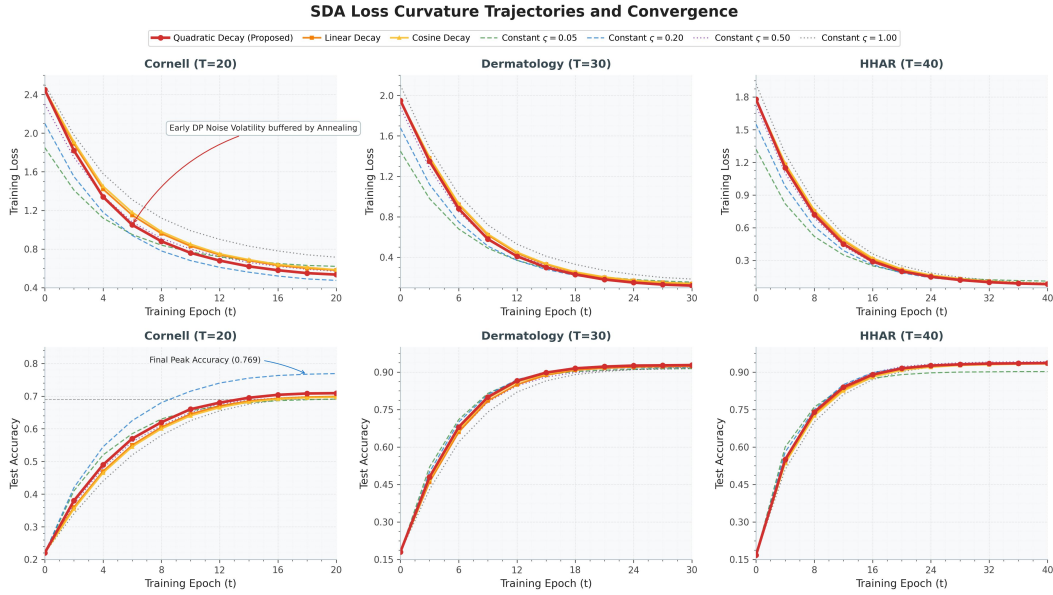
Table 7 reports all 12 cells of the component ablation. VIGN dominates (mean $\Delta = +0.109$), SDA contributes $+0.008$, and the offset is neutral. Macro-F1 tracks accuracy, confirming the effect is not an imbalance artifact.

C.2 CURVATURE ANNEALING SCHEDULE AND DYNAMICS

Figure 7 shows training-loss and test-accuracy trajectories under different SDA schedules at $\epsilon = 1$; Table 8 gives final accuracy by schedule. The quadratic anneal is the best decay shape and is competitive with the best fixed curvature without per-cell tuning.

Table 7: Full component ablation accuracy (mean $\pm$ std) across all $\epsilon \in \{1, 2, 4, 8\}$.

Data	ϵ	Full VIGN-SDA-PMSVM	w/o Offset	w/o SDA	w/o VIGN	Baseline PMSVM
Cornell	1	.701 $\pm$.031	.701 $\pm$.031	.695 $\pm$.023	.625 $\pm$.050	.617 $\pm$.053
Cornell	2	.784 $\pm$.020	.784 $\pm$.020	.733 $\pm$.034	.682 $\pm$.050	.683 $\pm$.050
Cornell	4	.831 $\pm$.013	.831 $\pm$.013	.790 $\pm$.010	.711 $\pm$.044	.713 $\pm$.042
Cornell	8	.835 $\pm$.020	.835 $\pm$.020	.814 $\pm$.010	.736 $\pm$.029	.736 $\pm$.027
ISOLET	1	.589 $\pm$.019	.590 $\pm$.018	.597 $\pm$.026	.443 $\pm$.023	.428 $\pm$.024
ISOLET	2	.769 $\pm$.018	.769 $\pm$.017	.758 $\pm$.029	.562 $\pm$.028	.557 $\pm$.027
ISOLET	4	.858 $\pm$.012	.858 $\pm$.013	.858 $\pm$.016	.645 $\pm$.026	.644 $\pm$.028
ISOLET	8	.889 $\pm$.005	.889 $\pm$.005	.906 $\pm$.009	.693 $\pm$.027	.698 $\pm$.025
USPS	1	.917 $\pm$.004	.916 $\pm$.003	.919 $\pm$.004	.871 $\pm$.012	.871 $\pm$.012
USPS	2	.924 $\pm$.004	.924 $\pm$.004	.927 $\pm$.005	.885 $\pm$.009	.886 $\pm$.009
USPS	4	.930 $\pm$.006	.929 $\pm$.005	.932 $\pm$.005	.897 $\pm$.007	.899 $\pm$.007
USPS	8	.932 $\pm$.006	.932 $\pm$.006	.934 $\pm$.005	.905 $\pm$.008	.907 $\pm$.008

Figure 7: Training loss (top) and test accuracy (bottom) for SDA schedules on Cornell, Dermatology, HHAR at $\epsilon = 1$. The proposed quadratic anneal buffers early DP-noise volatility and converges to a stable, high-accuracy solution.Table 8: Final accuracy by curvature schedule (mean $\pm$ std). “const= c ” disables annealing (fixed $\zeta \equiv c$).

Data	ϵ	quadratic	linear	cosine	const= .05	const= .2	const= .5	const= 1.0
Cornell	1	.709 $\pm$.041	.699	.697	.691	.769	.699	.699
Cornell	4	.839 $\pm$.009	.811	.819	.757	.825	.795	.701
Dermatology	1	.928 $\pm$.016	.923	.928	.914	.919	.914	.919
Dermatology	4	.955 $\pm$.039	.955	.955	.968	.959	.959	.959
HHAR	1	.936 $\pm$.005	.935	.933	.902	.936	.942	.938
HHAR	4	.960 $\pm$.001	.961	.960	.944	.962	.963	.956
Mean		.888	.881	.882	.863	.895	.879	.862

C.3 CLIPPING THRESHOLD R AND FEATURE NORMALIZATION

Table 9 reports each regime at its best R . Dropping per-sample L_2 normalization (norm=none) wins by +0.034 mean accuracy and makes accuracy robust to R ; unit-sphere mapping discards feature-magnitude signal and is sensitive to $R < \sqrt{2}$.

Table 9: norm=none (standard-scaled) vs. norm=l2 (unit sphere), each at its best clip R (VIGN on).

Data	ϵ	none: best R	none acc	l2: best R	l2 acc	$\Delta(\text{none}-\text{l2})$
Cornell	1	2.0	.743	0.1	.695	+.048
Cornell	4	0.5	.831	2.0	.807	+.024
Vehicle	1	0.25	.685	0.25	.658	+.027
Vehicle	4	1.0	.761	2.0	.749	+.012
Dermatology	1	0.1	.935	$\sqrt{2}$	.851	+.084
Dermatology	4	2.0	.962	0.5	.954	+.008

Table 10: Best accuracy per DP optimizer for VIGN (over all schedule/learning-rate settings).

Data	ϵ	DP-SGD	DP-Momentum	DP-Adam
HHAR	1	.864 $\pm$.009	.907 $\pm$.005	.934 $\pm$.004
HHAR	4	.876 $\pm$.005	.923 $\pm$.005	.959 $\pm$.004
ISOLET	1	.605 $\pm$.032	.612 $\pm$.031	.601 $\pm$.026
ISOLET	4	.827 $\pm$.020	.837 $\pm$.010	.862 $\pm$.012
USPS	1	.825 $\pm$.011	.889 $\pm$.009	.917 $\pm$.007
USPS	4	.850 $\pm$.011	.910 $\pm$.009	.931 $\pm$.005
Mean		.808	.846	.867

Table 11: Effect of linear learning-rate decay on the proposed gradient methods (accuracy; Diff = $|w/ - w/o|$).

Data	ϵ	GP	+decay	AGP	+decay
Cornell	1	.663	.673	.692	.692
Cornell	4	.771	.752	.772	.748
ISOLET	1	.336	.312	.339	.246
ISOLET	8	.789	.687	.814	.789
USPS	4	.927	.924	.926	.926

Table 12: Effect of Poisson batch size (accuracy, ISOLET).

ϵ	32	64	128	256	full
1	.059	.465	.501	.063	.037
2	.054	.614	.687	.074	.037
4	.124	.769	.804	.093	.105
8	.120	.834	.840	.157	.144

C.4 OPTIMIZER COMPATIBILITY

Table 10 gives the best accuracy per DP optimizer. DP-Adam is the strongest and most base-rate-robust choice for VIGN; learning-rate decay yields no benefit over a well-chosen constant rate.

C.5 LEARNING-RATE DECAY AND BATCH SIZE

Table 11 shows learning-rate decay gives no consistent advantage. Poisson batch size has a strong optimum: on this benchmark $B \in \{64, 128\}$ dominates, with very small or full batches degrading utility under DP noise (Table 12).

C.6 ROBUSTNESS TO CLASS IMBALANCE (NEGATIVE RESULT)

Table 13 sweeps the offset strength α under natural and synthetic imbalance. Across all regimes, α moves minority recall by ≤ 0.003 : the offset is applied and non-trivial in magnitude but is absorbed by DP-Adam’s second-moment normalization and VIGN’s re-normalization. We therefore set $\alpha = 0$ by default with no measurable loss.

C.7 CROSS-DATASET GRADIENT-NORMALIZATION PROBE

Table 14 gives the full numeric probe underlying Figure 6, extending the ISOLET analysis (Table 6) to USPS ($d=256$) and high-dimensional Cornell ($d=4,134$). The analytical feature-normalized

Table 13: Class-frequency margin offset: minority recall at best α vs. $\alpha = 0$. USPS uses minority classes subsampled to 10%/25%; ISOLET/Vehicle are natural.

Regime	Dataset	ϵ	best α	min-recall @best	min-recall @ $\alpha=0$	Δ
natural	ISOLET	4	0.1	.858	.857	+.001
natural	Vehicle	4	0.0	.744	.744	+.000
synthetic	USPS 10%	1	0.1	.468	.468	+.000
synthetic	USPS 25%	1	0.05	.779	.779	+.000

Table 14: Cross-dataset normalization probe (mean over five seeds). Near-0 residual % is the fraction of well-classified samples; $\max c_i$ is the peak per-sample reweight; the norm trend is the ratio $n_i(m=10)/n_i(m=1)$. VIGN is the only bounded, exactly violation-invariant scheme, and is decisively best in high dimensions (Cornell).

Data	ϵ	Normalization	Accuracy	Final clip %	Near-0 resid. %	$\max c_i$	Norm trend
ISOLET	1	Proposed VIGN	.589 $\pm$.019	41.5	65.1	1.4×10^4	1.00 $\times$
	1	L_2 deriv. ($1/\ v\ _2$)	.597 $\pm$.023	100.0	65.2	9.5×10^8	2.12 $\times$
	1	L_1 deriv. ($1/\ v\ _1$)	.596 $\pm$.025	100.0	65.4	2.0×10^8	0.74 $\times$
	4	Proposed VIGN	.858 $\pm$.012	11.1	91.0	1.4×10^4	1.00 $\times$
	4	L_2 deriv. ($1/\ v\ _2$)	.870 $\pm$.009	100.0	91.8	2.6×10^9	2.12 $\times$
	4	L_1 deriv. ($1/\ v\ _1$)	.871 $\pm$.011	100.0	91.9	5.4×10^8	0.74 $\times$
USPS	1	Proposed VIGN	.917 $\pm$.004	8.2	92.1	1.4×10^4	1.00 $\times$
	1	L_2 deriv. ($1/\ v\ _2$)	.916 $\pm$.005	100.0	92.0	7.2×10^{10}	2.24 $\times$
	1	L_1 deriv. ($1/\ v\ _1$)	.915 $\pm$.006	100.0	92.0	2.5×10^{10}	0.75 $\times$
Cornell ($d=4134$)	4	Proposed VIGN	.831 $\pm$.013	18.5	82.8	1.4×10^4	1.00 $\times$
	4	L_2 deriv. ($1/\ v\ _2$)	.725 $\pm$.039	100.0	74.1	2.0×10^{10}	1.87 $\times$
	4	L_1 deriv. ($1/\ v\ _1$)	.724 $\pm$.038	100.0	73.9	8.1×10^9	0.76 $\times$

gradient norm is $n_i = \|G_i\|_F / \sqrt{\|\Phi(x_i)\|_2^2 + 1}$; assuming m active violations ($v_{ik} \approx 1$), VIGN yields the exact invariant $n_i = \sqrt{2}$ (ratio 1.00 $\times$), L_2 grows as $\sqrt{m+1}$, L_1 under-corrects as $\sqrt{(m+1)/m}$, and standard clipping explodes as $\sqrt{m(m+1)}$. Empirically, VIGN keeps its per-sample weight bounded ($\max c_i \leq 1.4 \times 10^4$) so well-classified gradients decay and the final clip rate collapses; the naive derivative norms diverge ($\max c_i \sim 10^8$ – 10^{10}), remain 100% clipped, and collapse in high dimensions.